CM30370 Computer Science Final Project

Preliminary Report

Deep Learning-based Brease Cancer Detection System with Explainability

1. An Introduction

i. The project concept: Deep learning-based breast cancer detection system with explainability

   1. Explainability means providing the cause of breast cancer diagnosis for patients. If it is positive, the system explains why it is positive. If it is negative, this system also explains why.

ii. Project template

   1. Course: CM3015 Machine Learning and Neural Networks

   2. Idea Number: Project Idea Title 2 Deep Learning Breast Cancer Detection

iii. Motivation

   1. Doctor’s view

      a. Doctors want to create an Artificial Intelligence Breast Cancer Detection system to complement their jobs. Because they are suffering from burnouts [1], they need an AI breast cancer detection system which can help their job.

   2. Patient's view

      a. And patients also want the AI Breast Cancer Detection System. There aren’t enough doctors who can care for breast cancer patients. There is chronic shortage of doctors because human doctors must retire [2]. AI systems can complement shortage of doctors. And the co-work of the AI detection models and human doctors can enhance clinical ability. They can detect disease more accurately and faster than before.

   3. Why is explainability needed?

      a. That is because doctors must explain their diagnosis to their patients. If the detection system just provides diagnosis results without explanation, it is not the complete detection system. The diagnosis without explanation cannot be recognized as the real diagnosis by both patients and peer doctors.

2. A literature Review

i. Deep learning-based breast cancer detection in mammography [3]

   1. Summary

a. Breast cancer detection model using a modified EfficientNet V2 with enhanced attention mechanisms was proposed. This models classification and localization capabilities demonstrated robust performance.

2. What is good

   a. One of the state-of-the-art paper in the deep learning breast cancer detection models

   b. Model evaluation metrics are proposed. For example. AUROC (Area Under the Receiver Operating Characteristic Curve), Sensitivity, NPV (Negative Predictive Value), LLF (Lesion Localization Fraction)

   c. Image Classifier based on Encoder-Decoder Model with Transformer is proposed.

3. What to improve

   a. No Code is proposed.

ii. A curated mammography data set for use in computer-aided detection and diagnosis research [4]

   a. Summary

      i. Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM) is proposed

   b. What is good

      i. The problem which original DDSM dataset was solved with several image processing techniques.

      ii. The importance of standard dataset CBIS-DDSM is proposed

   c. What to improve

      i. Actual model research for this dataset is needed

iii. Deep Learning with Python [5]

   1. Summary

      a. Deep learning

         i. Deep learning is a subfield of machine learning. Deep refers to an idea of successive layers of representation.

      b. Mathematical building blocks of neural networks.

         i. Deep learning models are composed of the chain of tensors operations.

         ii. All tensor operations are differentiable

c. Introduction to tensor flow, pytorch, jax, keras

   i. Tensor flow, pytorch , jax and keras are popular AI frameworks.

d. Classification and regression

   i. The most common machine learning tasks are classification and regression.

e. Fundamentals of machine learning

   i. Generalization is the purpose of machine learning. Generalization is to perform accurately on never-before-seen inputs.

f. The universal workflow of machine learning

   i. Defining the problem -> develop a model -> deployment is the universial flow of machine learning

g. A deep dive on keras

   i. Keras offers a different workflow. Fit function is to train, evaluate function is to evaluate.

h. Image Classification

   i. ConvNet is good at computer vision tasks.

i. ConvNet Architecture Patterns

   i. Vision Transformer is an alternative to ConvNets for large computer vision datasets.

j. Interpret what ConvNets learn

   i. Grad-CAM visualize what areas are related with classification

k. Image Segmentation

   i. Image Segmentation is one of the important computer vision tasks.

l. Object Detection

   i. Object Detection locates objects within bounding boxes.

m. Timeseries Forecasting

   i. When the data have orders, recurrent networks are good.

n. Text classification

   i. Embedding is an efficient way to transform token ID info into latent space.

o. Language models and transformer

i. Transformer is a sequence modeling architecture that uses attention as the only mechanism to pass information across a sequence.

p. Text generation

i. Transformer architecture is an important ingredient of Large Language Models.

q. Image Generation

i. Image generation is done by latent spaces that capture statistical information about a image datasets.

r. Best practices for real world

i. An ensemble technique can often enhance model prediction ability

s. The future of AI

i. The evolution of model growing system can be an artificial general intelligence

t. Conclusions

i. Transformer uses attention to transform vector into another representation

2. What is Good

a. It is good for Comprehensive understanding of deep learning. ConvNet and Transformer are two important architectures in computer vision tasks.

b. It is good for Vision Transformer understanding.

c. Plenty of Coding examples which are very useful for project development

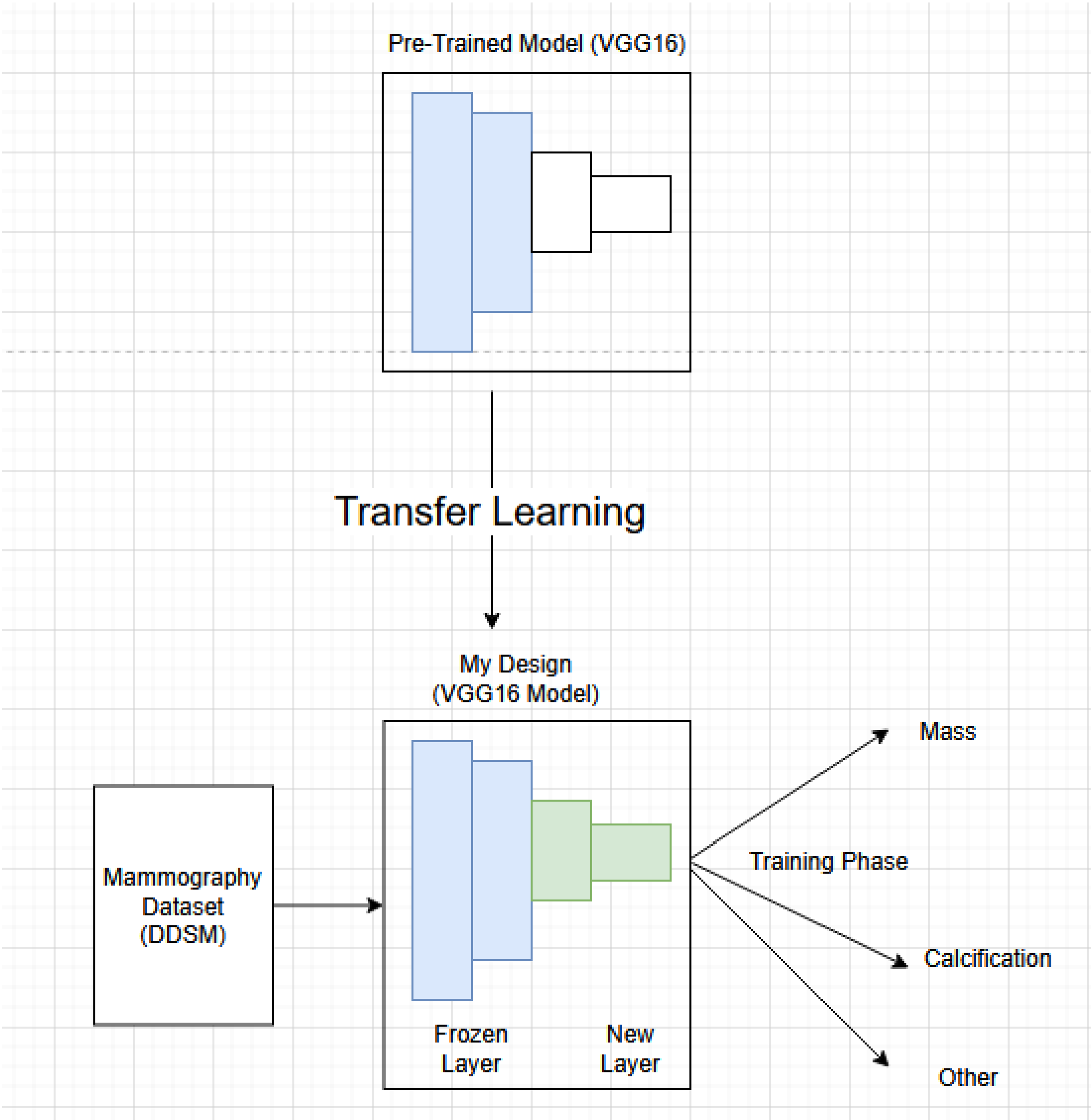
3. What to improve

a. This book's focus is on general deep learning, so medical appliance of deep learning is not much in this book

3. A Design

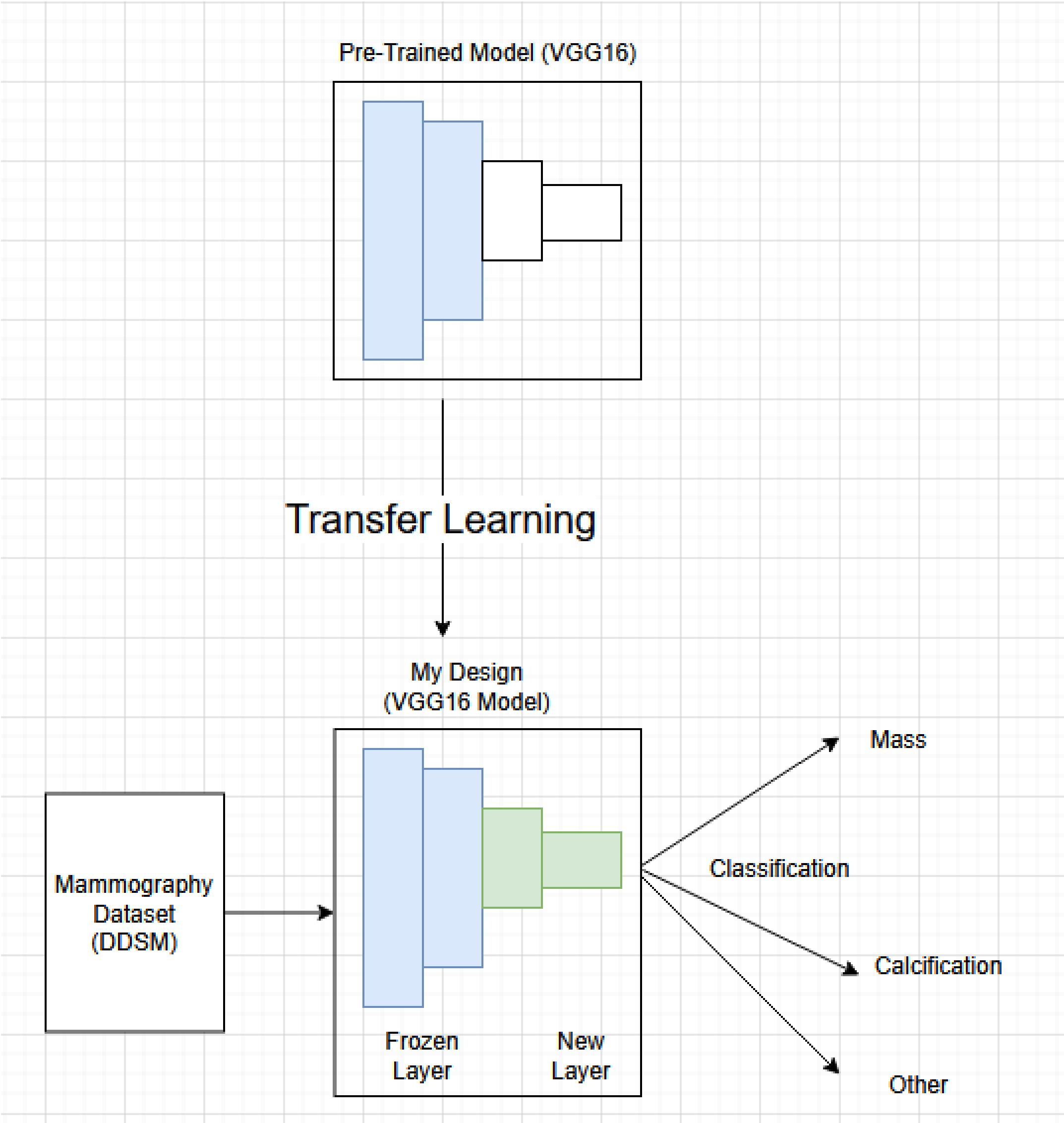
a. Diagram of Training Stage

i. My project design is based on transfer learning. And the fundamental CNN Model will be VGG16. in training stage, the model's frozen layers will be used which are from Pre-Trained Model. This pre-trained model is another VGG16 model trained with ImageNet or another huge medical datasets. So the frozen feature layers are from pre-trained model and new layers will be added to the frozen layer. So new layers of my new model will be trained with a Mammography dataset like DDSM. But frozen layers will not be changed because these layers are already trained from pre-trained model. The paper about transfer learning [6] will be studied for implementation

ii. 

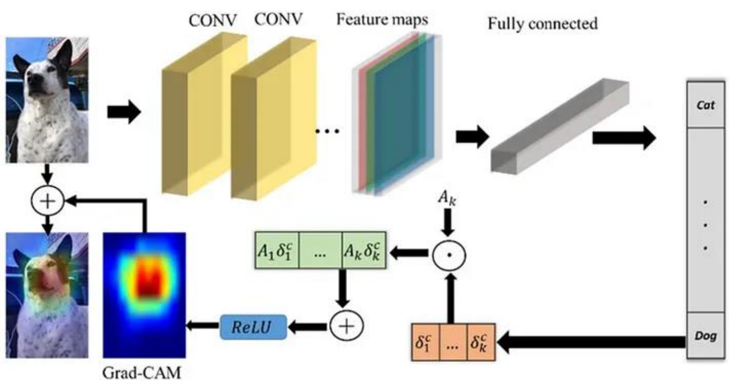
b. Diagram of Inference Stage

i. After the training stage, my new model will be used as a classifier. Having been trained with Hugh datasets (pre-training model's training stage) and mammography datasets (my model's training stage), the inference ability will be expected to be better than the model trained with only mammography datasets.

ii. 

c. Explainability

i. GradCAM technique will be used to gain explainability. This feature will be activated from Convolutional Neural Network's feature maps.

1. 

[7]

d. Workplan

Item	Due Date
Data Collection	December 19, 2025 ~ December 31, 2026
Model Architecture	January 1, 2026 ~ January 20, 2026

Development	
Model Training	January 21, 2026 ~ February 15, 2026
Model Evaluation	February 16, 2026 ~ March 10, 2026
Final Report	Mar 11, 20206 ~ March 23, 2026

4. A feature Prototype

a. Prototype architecture

i. My final goal is to make a breast cancer image classifier. So, my prototype is a simplified version of an image classifier. The neural network of this prototype is a simple Convolution layer architecture. In this stage, transfer learning is not used.

Prototype Architecture (Sequential Convolution Neural Network)

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 30, 30, 32)        896

 max_pooling2d (MaxPooling2D)  (None, 15, 15, 32)      0



 conv2d_1 (Conv2D)           (None, 13, 13, 128)       36992

 max_pooling2d_1 (MaxPooling2D)  (None, 6, 6, 128)     0



 conv2d_2 (Conv2D)           (None, 4, 4, 128)         147584

 flatten (Flatten)           (None, 2048)              0

 dense (Dense)               (None, 128)               262272

 dense_1 (Dense)             (None, 10)                1290

=================================================================
Total params: 449,034
Trainable params: 449,034
Non-trainable params: 0
```

b. Data for Prototype

i. for this prototype, cifar10 dataset is used. It is a simple dataset to check the model's classification ability. This dataset is not mammography data, but it is sufficient to check prototype model's classification ability.

c. Evaluation of the prototype

i. Accuray is used to evaluate the prototype

1. Accuracy = (number of correct predictions) / (total number of predictions)

ii. Result

1. Test Accuracy: 0.712

a. It is below 0.8 so not high. But it is prototype to experience image classifier's ability. So there are a lot of possibilities of improvement when the final model was used.

d. Prototype Code

i. Code is modified from this reference code [8]

Code

Step 0: import

import tensorflow as tf

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout

from tensorflow.keras.utils import to_categorical

import matplotlib.pyplot as plt

import numpy as np

Step 1: Load the CIFAR-10 dataset

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

Normalize(0-1)

x_train = x_train.astype('float32') / 255.0 x_test = x_test.astype('float32') / 255.0

Step 2: Build a image classifier ConvNet prototype model

```
def build_prototype():
    model = Sequential()
    model.add(Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)))
    model.add(MaxPooling2D((2, 2)))
    model.add(Conv2D(128, (3, 3), activation='relu'))
    model.add(MaxPooling2D((2, 2)))
    model.add(Conv2D(128, (3, 3), activation='relu'))
    model.add(Flatten()) # Flatten the 3D output to 1D for the Dense layers
    model.add(Dense(128, activation='relu'))
    model.add(Dense(10, activation='softmax')) # 10 output classes, use softmax for multi-class classification
    return model
model = build_prototype()
```

#Step 3: Compile the model

```
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
```

View the model summary

```
model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 30, 30, 32)        896
```

```
 max_pooling2d (MaxPooling2D  (None, 15, 15, 32)       0
 )


 conv2d_1 (Conv2D)           (None, 13, 13, 128)       36992


 max_pooling2d_1 (MaxPooling  (None, 6, 6, 128)        0
 2D)


 conv2d_2 (Conv2D)           (None, 4, 4, 128)         147584


 flatten (Flatten)           (None, 2048)              0


 dense (Dense)               (None, 128)               262272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=================================================================
Total params: 449,034
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```

#Step 4: Train the model

```
model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test), batch_size=128)
```

```
Epoch 1/10
391/391 [==============================] - 33s 84ms/step - loss: 1.6216 - accuracy: 0.4066 -
val_loss: 1.3845 - val_accuracy: 0.5081
Epoch 2/10
391/391 [==============================] - 33s 83ms/step - loss: 1.2484 - accuracy: 0.5551 -
val_loss: 1.1731 - val_accuracy: 0.5815
Epoch 3/10
391/391 [==============================] - 32s 83ms/step - loss: 1.0970 - accuracy: 0.6147 -
val_loss: 1.0538 - val_accuracy: 0.6291
Epoch 4/10
391/391 [==============================] - 33s 84ms/step - loss: 0.9979 - accuracy: 0.6498 -
val_loss: 0.9906 - val_accuracy: 0.6560
Epoch 5/10
391/391 [==============================] - 33s 84ms/step - loss: 0.9105 - accuracy: 0.6818 -
val_loss: 0.9409 - val_accuracy: 0.6702
Epoch 6/10
391/391 [==============================] - 33s 84ms/step - loss: 0.8437 - accuracy: 0.7067 -
val_loss: 0.9108 - val_accuracy: 0.6810
Epoch 7/10
391/391 [==============================] - 33s 84ms/step - loss: 0.7964 - accuracy: 0.7241 -
val_loss: 0.9573 - val_accuracy: 0.6663
Epoch 8/10
391/391 [==============================] - 34s 86ms/step - loss: 0.7368 - accuracy: 0.7410 -
val_loss: 0.8817 - val_accuracy: 0.6925
Epoch 9/10
391/391 [==============================] - 35s 90ms/step - loss: 0.6864 - accuracy: 0.7604 -
val_loss: 0.8361 - val_accuracy: 0.7158
Epoch 10/10
391/391 [==============================] - 48s 122ms/step - loss: 0.6446 - accuracy: 0.7756 -
val_loss: 0.8500 - val_accuracy: 0.7120
```

```
<keras.callbacks.History at 0x22b082c0910>
```

```
# Step 5: Evaluate the model

_, test_accuracy = model.evaluate(x_test, y_test, verbose=2)

print(f'Test accuracy: {test_accuracy}')
```

```
313/313 - 3s - loss: 0.8500 - accuracy: 0.7120 - 3s/epoch - 8ms/step
Test accuracy: 0.7120000123977661
```

5. Reference

i. https://www.rcr.ac.uk/news-policy/latest-updates/over-a-quarter-of-breast-cancer-doctors-set-to-retire-within-5-years-putting-patient-care-at-risk/

ii. https://www.rcr.ac.uk/news-policy/latest-updates/over-a-quarter-of-breast-cancer-doctors-set-to-retire-within-5-years-putting-patient-care-at-risk/

iii. Wang L. "Mammography with deep learning for breast cancer detection", Front Oncol. 2024 Feb, https://doi.org/10.3389/fonc.2024, 1281922

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v. Francoise Collet, "Deep Learning with Python", 2018, Manning, Shelter Island

vi. Tanjim Fatima, Hamdy Soliman, "Application of VGG16 Transfer Learning for Breast Cancer Detection", Information 2025, 16(3), 227; https://doi.org/10.3390/info16030227

vii. https://pub.towardsai.net/easy-explain-explainable-ai-with-gradcam-e187f85a697c

viii. https://akanshasaxena.com/challenge/deep-learning/day-8/